

Aussie Primary Academy | Australian Curriculum v9.0 | Learn, Investigate and Explain

Name: _____

Date: _____

LEARNING INTENTION	pose investigable questions, make reasoned predictions and plan a safe fair test
KEY IDEA	A fair test changes one variable, measures one outcome and keeps relevant conditions the same. A prediction states an expected result and a science reason.
WORKED MODEL	Use the inquiry cycle: question -> prediction -> safe method -> observations/data -> evidence-based conclusion -> evaluation -> communication.

SUCCESS CRITERIA - I CAN

☐ use the inquiry skill ☐ record evidence clearly ☐ evaluate quality ☐ explain a next step

A. BUILD THE SCIENCE IDEA

1. Turn 'plants and water' into a measurable investigable question.

2. Identify the changed, measured and three controlled variables in a fair test.

3. Improve this prediction: I think it will work.

B. APPLY AND EXPLAIN

4. Sequence six method steps so another group can repeat the test.

5. Add repeats, equipment, units and a safe handling rule.

6. Design a results table before collecting data.

C. INDEPENDENT CHECK

7. Audit a plan that changes water amount and light at the same time.

8. Rewrite the plan so the test is fair and the prediction can be checked.

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D. APPLY SCIENCE IN CONTEXT

9. Scenario: observe a pattern in shadow length and pose a question with a reasoned prediction

10. Write the question, prediction, variables, materials and numbered method.

11. Explain how repeats and controls make the evidence more trustworthy.

E. REASON, EVALUATE AND TRANSFER

12. Identify two safety or environmental responsibilities and the matching controls.

13. Swap plans with an imagined partner: list three details needed for exact repetition.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

[] I can pose investigable questions, make reasoned predictions and plan a safe fair test.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: _____

The most difficult science idea was _____ because _____

One correction or improvement I made: _____

A question I still have: _____

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Accept equivalent responses when the student uses accurate science ideas, relevant evidence and clear reasoning.

COMPLETE ANSWER KEY

1. Must name one changed factor and one measurable outcome.
2. Changed=independent; measured=dependent; controls stay constant.
3. Add expected direction and science reason.
- 4-6. Logical repeatable steps, suitable units, safety, repeats and labelled table.
7. Two variables prevent a clear cause.
8. Change one factor and standardise other conditions.
- 9-10. Accept a coherent, safe, measurable plan for the scenario.
11. Repeats test consistency; controls isolate the changed variable.
12. Examples: spills, heat, allergies, waste or habitat disturbance with specific controls.
13. Quantities, timing, equipment, measurement point and number of trials.

TEACHER CHECK AND NEXT STEP

[] Uses accurate vocabulary and a complete cause-and-effect explanation.

[] Supports conclusions with observations or data.

Next step: revisit one misconception, then transfer the idea to a new example.

COMMON MISCONCEPTION / WATCH OUT

A broad topic is not an investigable question.

A fair test does not mean every condition is identical; one chosen variable must change deliberately.

EXTENSION AND RE-TEACH

Re-teach: model one difficult item, complete one together, then ask the student to explain an independent correction.

Extension: apply the science idea to a new real-world example and justify the explanation with observations or data.